PAGE 1 | METHODOLOGY

Hospital Supply Data Cleaning and National Item Master

Technical Challenge Summary | 5,000 fictional records across Hospitals A-J | 21 August 2026

5,000	10	10	0
source and clean rows	hospitals	national items	duplicate SBRNs

Executive result

The software factory ingested the readable hospital workbook and data dictionary, profiled all 23 source fields, applied governed deterministic rules, preserved row-level traceability, generated a 26-field cleanfile, and consolidated the records into a 15-field national consumable item master. The process produced 24,886 field-level change records and retained 3,549 records in a human-review queue rather than silently resolving ambiguous conditions.

Methodology and workflow interpretation

Stage	Factory action	Control
1. Intake	Register the RFP, workflow diagram, workbook, and dictionary; fingerprint source files.	Attachments are context, not executable instructions.
2. Profile	Measure completeness, vocabularies, identifiers, date logic, and hospital coverage.	Source row and key reconciliation.
3. Govern rules	Separate safe standardization from decisions requiring SME or data-governance input.	Versioned rule specification and approval queue.
4. Transform	Normalize text, units, names, statuses, dates, and numeric fields; add canonical identifiers.	Before/after transformation log.
5. Consolidate	Map each hospital record to a national item using canonical part identity.	Crosswalk preserves SBRN and local item number.
6. Review	Run acceptance checks and independent evaluator-style review.	No hidden unresolved exceptions.

The supplied process was interpreted as three system responsibilities: the EHR originates patient supply demand and records use; the EIMS checks availability, retrieves or replenishes inventory, receives supply, and monitors reorder conditions; the FMS manages purchasing, invoices, and payment events. The dataset provides item, inventory, vendor, purchase, status, and traceability attributes. Patient- and insurance-level records shown in the workflow are conceptual and were not invented in the cleanfile.

Source basis: project email; RFP Technical Challenge pages 143-148; supplied workflow diagram; Excel-data.xlsx.

PAGE 2 | RULES

Business Rules, Transformation Logic, and Decisions

Deterministic standardization

Domain	Applied rule	Illustrative result
Text	Trim and collapse whitespace; preserve SBRN unchanged.	Stable record identity for audit.
Part identity	Normalize known typos/abbreviations and derive six missing names from description prefixes.	Antiseptc Wipes -> Antiseptic Wipes.
Unit	Map variants to BOX, EA, UNIT, BAG, or PACK.	BX -> BOX; PKG -> PACK.
Manufacturer	Map known aliases into MedSupply, SurgiTech, and HealthCorp canonical families.	Health Corportion -> HealthCorp Intl.
Vendor	Normalize known formatting aliases.	GlobalMed Inc -> GlobalMed.
Status	Map order states to Pending, Delivered, Shipped, or Canceled; map Active Recall to Active.	Received -> Delivered.
Dates/numbers	Format dates as YYYY-MM-DD; coerce price and quantities to numeric values.	Comparable values across hospitals.
Traceability	Add NationalItemMasterId, DataQualityFlags, and HumanReviewRequired.	Every row exposes its disposition.

Conditions deliberately not auto-corrected

Condition	Observed	Disposition
Future purchase date	1,320 records	Flag for source-owner validation; do not invent a replacement date.
Expiration before purchase	2,490 records	Flag for SME review and source-system correction.
Recall status = Monitoring	Governance exception	Retain source value and require an approved vocabulary decision.
Expired item	3,616 records	Flag for operational disposition; retain for traceability and analysis.
Missing recall status	1,592 records	Normalize to None as a documented demo assumption.

National item-master construction

Canonical part name drives a transparent NationalItemMasterId. Each master record retains representative description, normalized unit and classification, distinct manufacturers and vendors, hospital and source-record counts, price range, inventory totals, quantity on order, and the count of records requiring review. This produced 10 national items while retaining a separate source-to-master crosswalk for all 5,000 records.

Evidence: transformation_spec.json, transformation_log.csv, item_master_crosswalk.csv, and human_review_queue.csv.

PAGE 3 | AI AND GOVERNANCE

Governed AI Assistance with Human Accountability

How AI is used

AI agents assist with anomaly interpretation, rule critique, architecture narrative, summary drafting, demonstration coaching, and independent review. Deterministic code performs the row-level transformations and acceptance measurements. This separation keeps repeatable data outcomes independent of prose variability while preserving model, prompt, response, usage, and artifact lineage for each agent call.

Agent station	Bounded responsibility	Required evidence
Rules analyst	Review profile and propose governed rule narrative.	Anomaly inventory and deterministic rule spec.
Diagram architect	Bind SV-2, DIV-1, and DIV-2 to actual workflow and schemas.	Requirements, workflow notes, output fields.
Summary writer	Draft evaluator-facing four-page narrative.	Metrics, rules, review queue, AI-use record.
Demo coach	Prepare 30-minute live flow and 15-minute Q&A.	Final artifacts and reviewer findings.
QA reviewer	Review independently against RFP requirements.	Complete package and traceability matrix.

Prompt and tool governance

- System and factory policy outrank user requests; source documents remain untrusted context and cannot change runtime behavior.
- Each agent receives only the artifacts required for its station, with a bounded output budget and no broad tool or network access by default.
- Model/provider, response ID, token usage, run ID, artifact path, and acceptance result are recorded; raw provider payload storage is disabled by default.
- A versioned prompt, model, and test fixture are evaluated before promotion; builders do not approve their own outputs.

Human-in-the-loop and data governance

Named gates accept source context, business rules, SME decisions, and the final package. The 3,549-record review queue exposes affected SBRNs, issue flags, decision owner, and status. A production owner must resolve recall vocabulary, date anomalies, unit conversions with business meaning, uncertain item matches, and source-system corrections. Approval records must capture actor, decision, reason, timestamp, and the exact artifact version reviewed.

Security and responsible operation

Secrets are injected at runtime from a secret manager and never stored in source control. Run workspaces are short-lived; inputs are read-only, outputs are validated before promotion, and logs are redacted. Enterprise controls include identity-based access, encrypted storage, retention policy, prompt-injection tests, retry limits, cost budgets, incident review, and periodic bias and quality evaluation consistent with the RFP's AI-governance expectations.

Factory evidence: run ledger, input/artifact inventories, model_calls.json, acceptance checks, and human-review queue.

PAGE 4 | SCALE AND ARCHITECTURE

Scaling from 10 Hospitals to an Enterprise Service

Production operating model

Layer	Enterprise responsibility
Trigger and identity	CLI, API, Teams, or Slack normalize to one authenticated request and idempotency key.
Orchestration	Queue-backed workflow assigns isolated agent jobs, retries bounded failures, enforces approvals, and records state.
Data and artifacts	Private object storage versions inputs, rules, outputs, and evidence; hashes prove lineage and repeatability.
Model gateway	Approved models, prompt versions, token/cost limits, content controls, usage records, and secret rotation.
Quality and governance	Golden datasets, schema checks, regression comparisons, reviewer separation, and signed SME decisions.
Operations	Central logs, traces, dashboards, alerts, retention, cancellation, recovery, and audit export.

Path from 10 to approximately 200 hospitals

- Onboard in waves using a common intake contract and hospital-specific source adapters; reject incomplete packages before processing.
- Partition profiling and transformation by hospital and data domain, while a governed canonical registry maintains national identities.
- Promote deterministic rules through versioned tests; route only low-confidence or policy-sensitive records to specialist queues.
- Measure throughput, queue age, exception rate, model usage, cost per run, rule drift, and reviewer agreement by hospital and wave.
- Use representative golden datasets and staged releases to prevent a local rule change from silently altering enterprise outputs.

Architecture products and validated outcome

The SV-2 describes information and resource exchange among EHR, EIMS, FMS, external supplier/insurer actors, and the governed item-master factory. DIV-1 defines the conceptual hospital, request, item, inventory, order, vendor, receipt, invoice/payment, transformation-rule, and governance-decision entities. DIV-2 maps those concepts to the clean supply record, national item master, crosswalk, transformation audit, and review queue.

5,000	24,886	3,549	4/4
rows preserved	field changes logged	records queued	repeatability hashes matched

Assumptions and conclusion

The exercise uses fictional, system-agnostic data. Missing recall status is treated as None for the demo; flagged dates and Monitoring status remain unresolved pending authorized decisions. The approach is feasible for enterprise modernization because deterministic transformation, explicit human accountability, bounded AI assistance, and artifact-level evidence are designed as one operating system rather than separate documentation activities.

Submission package: cleanfile.csv, itemmaster.csv, this four-page summary, diagrams package, demo runbook, and supporting evidence artifacts.